RUNWAY MANAGEMENT SYSTEM

this project is based on runway management where multiple planes either land or take off in a controlled manner using multithread concept in C.

Coding:

```
#include<stdio.h>
#include<pthread.h>
#include<stdlib.h>
#include<unistd.h>
pthread_mutex_t runway;
int no_planes;
int flag=0;
void *rway(void *plane)
{
   int *plane_no = (int*)plane;
   if(*plane_no <= (no_planes)/2)
   {
      printf("plane %d is landing\n",*plane_no);
      pthread_mutex_lock(&runway);
      flag = 1;
      sleep(1);
      printf("plane %d is landed and runway is clear now\n",*plane_no);
      flag = 0;
      pthread_mutex_unlock(&runway);
```

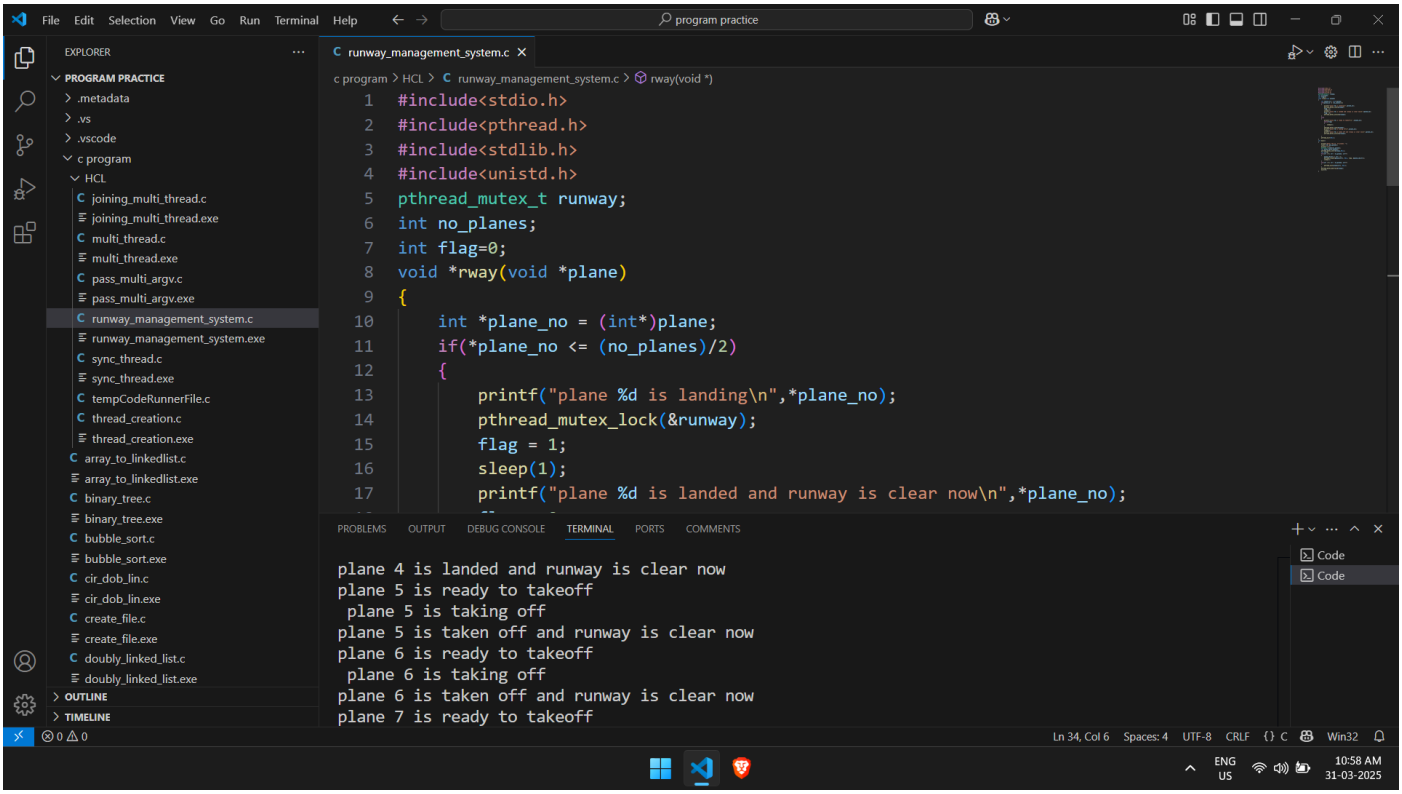
```
    }
    else
    {
        printf("plane %d is ready to takeoff\n ",*plane_no);
        while(flag)
        {
            sleep(2);
        }
        pthread_mutex_lock(&runway);
        printf("plane %d is taking off\n",*plane_no);
        sleep(1);
        printf("plane %d is taken off and runway is clear now\n",*plane_no);
        pthread_mutex_unlock(&runway);

    }
    pthread_exit(NULL);
}
int main()
{
    printf("Enter the no. of planes: ");
    scanf("%d",&no_planes);
    printf("\n");
    pthread_t plane[no_planes];
    int plane_id[no_planes];
    pthread_mutex_init(&runway,NULL);
```

```
    int itr=0;
    for(itr = 0; itr < no_planes; itr++)
    {
        plane_id[itr] = itr + 1;
        pthread_create(&plane[itr], NULL, rway, &plane_id[itr]);
        sleep(1);
    }
    for(itr = 0; itr < no_planes; itr++)
    {
        pthread_join(plane[itr], NULL);
    }
    pthread_mutex_destroy(&runway);
    exit(0);
}
```

Assumptions:

- The first half of the planes attempt to land.
- The remaining attempt to take off.

Output:


File Edit Selection View Go Run Terminal Help
program practice
EXPLORER
PROGRAM PRACTICE
.metadata
.vs
.vscode
c program
HCL
joining_multi_thread.c
joining_multi_thread.exe
multi_thread.c
multi_thread.exe
pass_multi_argv.c
pass_multi_argv.exe
runway_management_system.c
runway_management_system.exe
sync_thread.c
sync_thread.exe
tempCodeRunnerFile.c
thread_creation.c
thread_creation.exe
array_to_linkedlist.c
array_to_linkedlist.exe
binary_tree.c
binary_tree.exe
bubble_sort.c
bubble_sort.exe
cir_dob_lin.c
cir_dob_lin.exe
create_file.c
create_file.exe
doubly_linked_list.c
doubly_linked_list.exe
OUTLINE
TIMELINE
runway_management_system.c
c program > HCL > runway_management_system.c > rway(void *)
1 #include<stdio.h>
2 #include<pthread.h>
3 #include<stdlib.h>
4 #include<unistd.h>
5 pthread_mutex_t runway;
6 int no_planes;
7 int flag=0;
8 void *rway(void *plane)
9 {
10 int *plane_no = (int*)plane;
11 if(*plane_no <= (no_planes)/2)
12 {
13 printf("plane %d is landing\n",*plane_no);
14 pthread_mutex_lock(&runway);
15 flag = 1;
16 sleep(1);
17 printf("plane %d is landed and runway is clear now\n",*plane_no);
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
plane 4 is landed and runway is clear now
plane 5 is ready to takeoff
plane 5 is taking off
plane 5 is taken off and runway is clear now
plane 6 is ready to takeoff
plane 6 is taking off
plane 6 is taken off and runway is clear now
plane 7 is ready to takeoff
Code
Code
Ln 34, Col 6 Spaces: 4 UTF-8 CRLF C Win32
ENG US
10:58 AM 31-03-2025